

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler
```

```
In [2]: df=pd.read_csv("/home/mostafashazly/Downloads/Large-Scale Financial Fraud Da
print(df.head())
```

```
      organization transaction_id      user_id user_account_age_day
s \
0  FinovaTech_000    TXN00000044  FinovaTech_000_U0000    167
0
1  FinovaTech_000    TXN00000004  FinovaTech_000_U0000    167
0
2  FinovaTech_000    TXN00000033  FinovaTech_000_U0000    167
0
3  FinovaTech_000    TXN00000050  FinovaTech_000_U0000    167
0
4  FinovaTech_000    TXN00000047  FinovaTech_000_U0000    167
0

      kyc_verified payment_method card_type  transaction_amount currency \
0              1          bkash    debit          1553.81      BDT
1              1          bkash    debit           136.49      BDT
2              1           card    credit          3432.40      BDT
3              1          bkash    credit           503.59      BDT
4              0           bank    credit          1908.35      BDT

      fee_amount  ... device_type operating_system browser merchant_category \
0         31.08  ...      mobile          Windows   Safari          fashion
1          2.73  ...     desktop          Android    Edge          grocery
2         68.65  ...     tablet             iOS    Safari    electronics
3         10.07  ...     tablet          Windows    Chrome          fashion
4         38.17  ...     tablet          Windows    Chrome          grocery

      transaction_type otp_used is_fraud hour is_night  time_diff
0         purchase          1         0     8         0         0.0
1      withdrawal          0         0     6         0    1719608.0
2         transfer          1         0    19         0    827250.0
3         purchase          1         0    19         0   1035744.0
4         transfer          1         0    12         0   408465.0
```

[5 rows x 24 columns]

```
In [3]: print(df.columns.to_list())
```

```
['organization', 'transaction_id', 'user_id', 'user_account_age_days', 'kyc_
verified', 'payment_method', 'card_type', 'transaction_amount', 'currency',
'fee_amount', 'transaction_timestamp', 'day_of_week', 'city', 'country', 'de
vice_type', 'operating_system', 'browser', 'merchant_category', 'transaction
_type', 'otp_used', 'is_fraud', 'hour', 'is_night', 'time_diff']
```

```
In [4]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   organization                          1000000 non-null  str
1   transaction_id                        1000000 non-null  str
2   user_id                              1000000 non-null  str
3   user_account_age_days                1000000 non-null  int64
4   kyc_verified                         1000000 non-null  int64
5   payment_method                       1000000 non-null  str
6   card_type                            1000000 non-null  str
7   transaction_amount                   1000000 non-null  float64
8   currency                             1000000 non-null  str
9   fee_amount                           1000000 non-null  float64
10  transaction_timestamp                 1000000 non-null  str
11  day_of_week                           1000000 non-null  str
12  city                                 1000000 non-null  str
13  country                              1000000 non-null  str
14  device_type                          1000000 non-null  str
15  operating_system                     1000000 non-null  str
16  browser                              1000000 non-null  str
17  merchant_category                    1000000 non-null  str
18  transaction_type                      1000000 non-null  str
19  otp_used                             1000000 non-null  int64
20  is_fraud                             1000000 non-null  int64
21  hour                                 1000000 non-null  int64
22  is_night                             1000000 non-null  int64
23  time_diff                            1000000 non-null  float64
dtypes: float64(3), int64(6), str(15)
memory usage: 312.0 MB

```

```

In [5]: fraud_rate = df["is_fraud"].mean()
        print(fraud_rate)

```

```
0.09171
```

```

In [6]: accounts_not_verified = df[(df['kyc_verified']==0) & (df['is_fraud']== 1)]
        print(accounts_not_verified)

```

	organization	transaction_id	user_id	\
190	FinovaTech_000	TXN00000196	FinovaTech_000_U0003	
353	FinovaTech_000	TXN00000367	FinovaTech_000_U0007	
588	FinovaTech_000	TXN00000551	FinovaTech_000_U0011	
616	FinovaTech_000	TXN00000625	FinovaTech_000_U0012	
621	FinovaTech_000	TXN00000636	FinovaTech_000_U0012	
...	...	...	...	
999310	FinovaTech_199	TXN00999315	FinovaTech_199_U0086	
999570	FinovaTech_199	TXN00999586	FinovaTech_199_U0091	
999810	FinovaTech_199	TXN00999829	FinovaTech_199_U0096	
999821	FinovaTech_199	TXN00999827	FinovaTech_199_U0096	
999850	FinovaTech_199	TXN00999860	FinovaTech_199_U0097	

	user_account_age_days	kyc_verified	payment_method	card_type	\
190	187	0	bkash	credit	
353	34	0	bkash	credit	
588	1123	0	card	credit	
616	1156	0	bkash	credit	
621	1156	0	card	debit	
...	...	...	...	...	
999310	1109	0	bank	credit	
999570	779	0	nagad	credit	
999810	1206	0	card	credit	
999821	1206	0	bkash	debit	
999850	862	0	card	debit	

	transaction_amount	currency	fee_amount	...	device_type	\
190	402.67	BDT	8.05	...	desktop	
353	698.01	BDT	13.96	...	tablet	
588	418.29	BDT	8.37	...	tablet	
616	2224.75	BDT	44.49	...	tablet	
621	817.26	BDT	16.35	...	mobile	
...	...	...	...	...	...	
999310	801.01	BDT	16.02	...	desktop	
999570	3280.25	BDT	65.61	...	desktop	
999810	2808.73	BDT	56.17	...	tablet	
999821	2338.01	BDT	46.76	...	mobile	
999850	179.87	BDT	3.60	...	desktop	

	operating_system	browser	merchant_category	transaction_type	otp_used
190	iOS	Safari	travel	withdrawal	0
353	Android	Chrome	travel	withdrawal	0
588	iOS	Edge	electronics	transfer	0
616	iOS	Chrome	fashion	withdrawal	0
621	Android	Edge	fashion	withdrawal	0
...	...	...	...	...	...
999310	Android	Edge	grocery	withdrawal	0
999570	Android	Chrome	fashion	transfer	0
999810	Android	Safari	electronics	withdrawal	0
999821	Android	Edge	fashion	transfer	0
999850	Android	Edge	travel	transfer	0

	is_fraud	hour	is_night	time_diff
190	1	4	1	1160043.0
353	1	3	1	674345.0

588	1	2	1	280261.0
616	1	4	1	723124.0
621	1	1	1	2088692.0
...	...	...	...	...
999310	1	0	1	135707.0
999570	1	2	1	327296.0
999810	1	2	1	2309772.0
999821	1	2	1	816367.0
999850	1	3	1	0.0

[5640 rows x 24 columns]

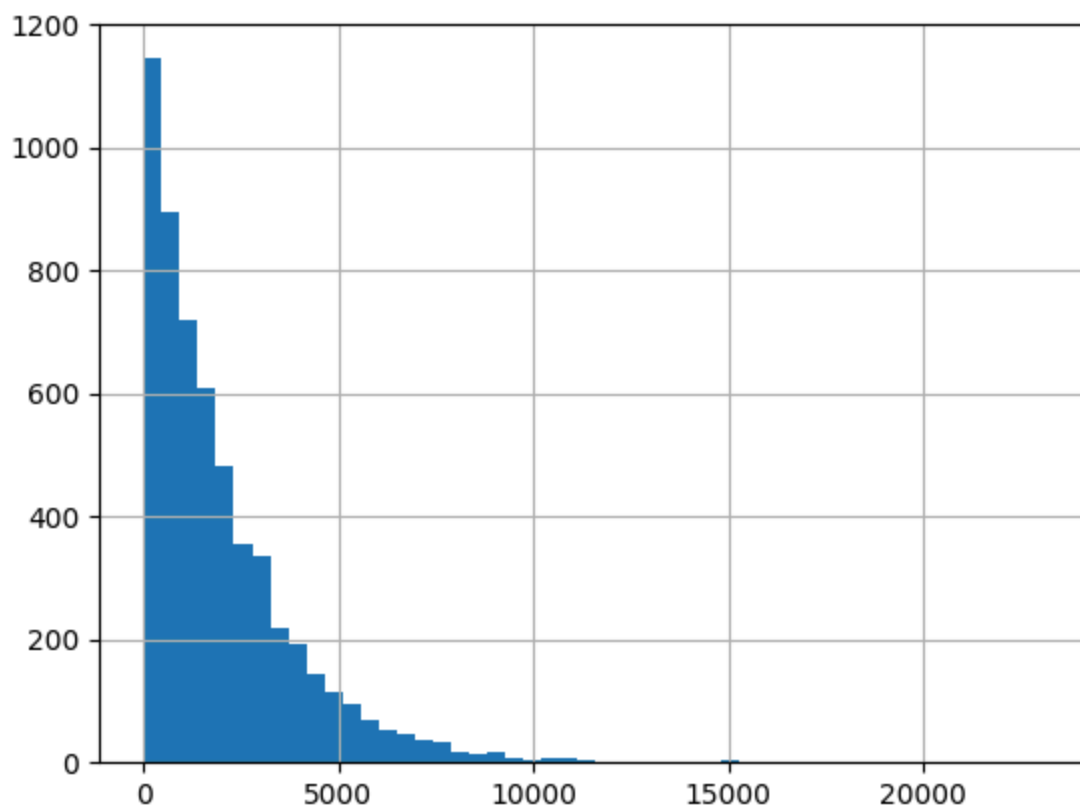
```
In [7]: number_of_duplicats = accounts_not_verified.duplicated().sum()
number_of_missing_values = accounts_not_verified.isna().sum()
print(f"number_of_duplicats is: {number_of_duplicats}\n")
print(f"number_of_missing_values is: {number_of_missing_values}")
```

number_of_duplicats is: 0

```
number_of_missing_values is: organization      0
transaction_id      0
user_id              0
user_account_age_days  0
kyc_verified         0
payment_method       0
card_type            0
transaction_amount    0
currency             0
fee_amount           0
transaction_timestamp 0
day_of_week          0
city                 0
country              0
device_type          0
operating_system     0
browser              0
merchant_category    0
transaction_type      0
otp_used             0
is_fraud             0
hour                 0
is_night             0
time_diff            0
dtype: int64
```

```
In [8]: accounts_not_verified['transaction_amount'].mean()
accounts_not_verified['transaction_amount'].hist(bins=50)
```

Out[8]: <Axes: >



```
In [9]: accounts_not_verified['transaction_amount'].nlargest(10)
```

```
Out[9]: 200944    23154.39
192873    17657.30
788979    17410.96
280996    17249.32
116451    16016.37
653428    15957.14
151377    15726.68
133250    15427.95
180865    15169.26
245406    15100.40
Name: transaction_amount, dtype: float64
```

```
In [10]: print("Fraud Mean:", accounts_not_verified['transaction_amount'].mean())

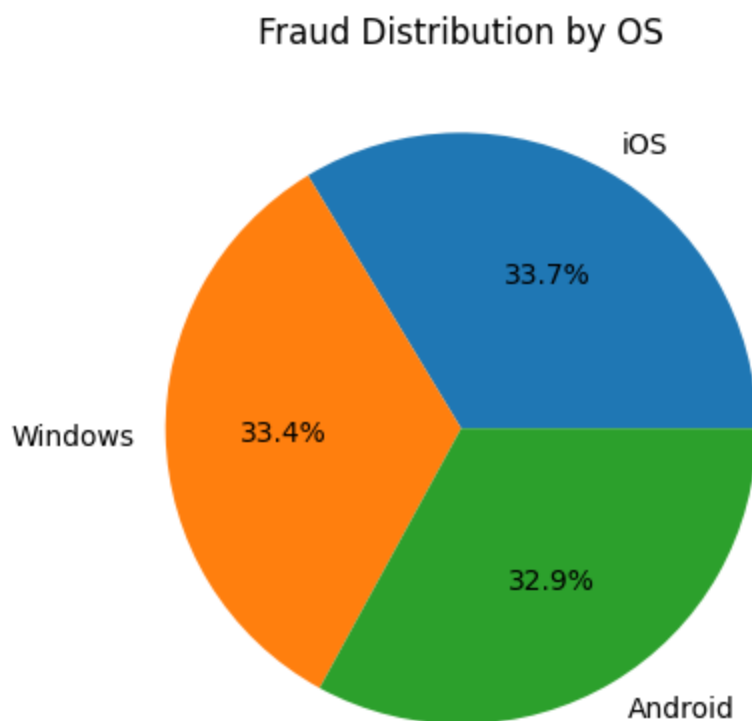
normal_ops = df[df['is_fraud'] == 0]
print("Normal Mean:", normal_ops['transaction_amount'].mean())
```

```
Fraud Mean: 1998.8218333333334
Normal Mean: 2001.396133536646
```

```
In [11]: fraud_by_os = accounts_not_verified['operating_system'].value_counts()
print(fraud_by_os)
```

```
operating_system
iOS          1899
Windows      1883
Android      1858
Name: count, dtype: int64
```

```
In [12]: fraud_by_os.plot(kind='pie', autopct='%1.1f%%')  
plt.title("Fraud Distribution by OS")  
plt.show()
```



```
In [13]: print(df.groupby('browser')['is_fraud'].mean(), "\n")  
print(df.groupby('card_type')['is_fraud'].mean(), "\n")  
print(df.groupby('currency')['is_fraud'].mean(), "\n")  
print(df.groupby('merchant_category')['is_fraud'].mean(), "\n")  
print(df.groupby('device_type')['is_fraud'].mean(), "\n")
```

```
browser
Chrome    0.091917
Edge      0.091536
Safari    0.091678
Name: is_fraud, dtype: float64
```

```
card_type
credit    0.091503
debit     0.091916
Name: is_fraud, dtype: float64
```

```
currency
BDT       0.09171
Name: is_fraud, dtype: float64
```

```
merchant_category
electronics    0.090059
fashion        0.092329
grocery        0.092143
travel         0.092311
Name: is_fraud, dtype: float64
```

```
device_type
desktop    0.091681
mobile     0.091516
tablet     0.091934
Name: is_fraud, dtype: float64
```

```
In [14]: print(df.groupby('transaction_type')['is_fraud'].mean(), "\n")
print(df.groupby('is_night')['is_fraud'].mean(), "\n")
print(df.groupby('payment_method')['is_fraud'].mean(), "\n")
```

```
transaction_type
purchase    0.091576
transfer    0.091688
withdrawal  0.091867
Name: is_fraud, dtype: float64
```

```
is_night
0    0.006105
1    0.299754
Name: is_fraud, dtype: float64
```

```
payment_method
bank        0.091190
bkash       0.091986
card        0.092270
nagad       0.091394
Name: is_fraud, dtype: float64
```

```
In [15]: print(df.groupby('currency')['is_fraud'].mean(), "\n")
print(df.groupby('country')['is_fraud'].mean(), "\n")
print(df.groupby('city')['is_fraud'].mean(), "\n")
print(df.groupby('time_diff')['is_fraud'].mean(), "\n")
```

```

currency
BDT    0.09171
Name: is_fraud, dtype: float64

```

```

country
Bangladesh    0.09171
Name: is_fraud, dtype: float64

```

```

city
Chittagong    0.092141
Dhaka         0.091575
Khulna        0.091091
Rajshahi      0.092032
Name: is_fraud, dtype: float64

```

```

time_diff
0.0          0.297850
2.0          1.000000
3.0          0.333333
4.0          0.000000
6.0          0.000000
...
14120645.0   0.000000
14370376.0   0.000000
14641465.0   0.000000
14736916.0   0.000000
17661552.0   0.000000
Name: is_fraud, Length: 817408, dtype: float64

```

```

In [16]: col_drop = [
            'currency', 'country', 'city',
            'browser', 'device_type', 'card_type',
            'transaction_id', 'user_id', 'organization',
            'transaction_timestamp'
        ]

df_clean = df.drop(columns=col_drop)
print(df_clean.head())

```


	user_account_age_days	kyc_verified	payment_method	transaction_amount	\
0	1670	1	bkash	1553.81	
1	1670	1	bkash	136.49	
2	1670	1	card	3432.40	
3	1670	1	bkash	503.59	
4	1670	0	bank	1908.35	

	fee_amount	day_of_week	operating_system	merchant_category	transaction_type
0	31.08	Friday	Windows	fashion	purchase
1	2.73	Thursday	Android	grocery	withdrawal
2	68.65	Saturday	iOS	electronics	transfer
3	10.07	Thursday	Windows	fashion	purchase
4	38.17	Tuesday	Windows	grocery	transfer

	otp_used	is_fraud	hour	is_night	time_diff
0	1	0	8	0	0.0
1	0	0	6	0	1719608.0
2	1	0	19	0	827250.0
3	1	0	19	0	1035744.0
4	1	0	12	0	408465.0

```
In [17]: df_final = pd.get_dummies(df_clean)

print(df_final.head())
```

	user_account_age_days	kyc_verified	transaction_amount	fee_amount	\
0	1670	1	1553.81	31.08	
1	1670	1	136.49	2.73	
2	1670	1	3432.40	68.65	
3	1670	1	503.59	10.07	
4	1670	0	1908.35	38.17	

	otp_used	is_fraud	hour	is_night	time_diff	payment_method_bank	...
\							
0	1	0	8	0	0.0	False	...
1	0	0	6	0	1719608.0	False	...
2	1	0	19	0	827250.0	False	...
3	1	0	19	0	1035744.0	False	...
4	1	0	12	0	408465.0	True	...

	operating_system_Android	operating_system_Windows	operating_system_iOS
\			
0	False	True	False
1	True	False	False
2	False	False	True
3	False	True	False
4	False	True	False

	merchant_category_electronics	merchant_category_fashion	\
0	False	True	
1	False	False	
2	True	False	
3	False	True	
4	False	False	

	merchant_category_grocery	merchant_category_travel	\
0	False	False	
1	True	False	
2	False	False	
3	False	False	
4	True	False	

	transaction_type_purchase	transaction_type_transfer	\
0	True	False	
1	False	False	
2	False	True	
3	True	False	
4	False	True	

	transaction_type_withdrawal
0	False
1	True
2	False
3	False
4	False

[5 rows x 30 columns]

```
In [18]: scaler = MinMaxScaler(feature_range=(0, 1))
```

```
In [19]: df_scaled = pd.DataFrame(scaler.fit_transform(df_final), columns=df_final.columns)
print(df_scaled.head())
```

```

   user_account_age_days  kyc_verified  transaction_amount  fee_amount \
0          0.834171         1.0         0.049988      0.049994
1          0.834171         1.0         0.004391      0.004391
2          0.834171         1.0         0.110424      0.110427
3          0.834171         1.0         0.016201      0.016198
4          0.834171         0.0         0.061394      0.061398

   otp_used  is_fraud      hour  is_night  time_diff  payment_method_bank \
0         1.0       0.0  0.347826        0.0   0.000000                0.0
1         0.0       0.0  0.260870        0.0   0.097364                0.0
2         1.0       0.0  0.826087        0.0   0.046839                0.0
3         1.0       0.0  0.826087        0.0   0.058644                0.0
4         1.0       0.0  0.521739        0.0   0.023127                1.0

   ...  operating_system_Android  operating_system_Windows \
0  ...                        0.0                        1.0
1  ...                        1.0                        0.0
2  ...                        0.0                        0.0
3  ...                        0.0                        1.0
4  ...                        0.0                        1.0

   operating_system_iOS  merchant_category_electronics \
0                   0.0                        0.0
1                   0.0                        0.0
2                   1.0                        1.0
3                   0.0                        0.0
4                   0.0                        0.0

   merchant_category_fashion  merchant_category_grocery \
0                   1.0                        0.0
1                   0.0                        1.0
2                   0.0                        0.0
3                   1.0                        0.0
4                   0.0                        1.0

   merchant_category_travel  transaction_type_purchase \
0                   0.0                        1.0
1                   0.0                        0.0
2                   0.0                        0.0
3                   0.0                        1.0
4                   0.0                        0.0

   transaction_type_transfer  transaction_type_withdrawal
0                   0.0                        0.0
1                   0.0                        1.0
2                   1.0                        0.0
3                   0.0                        0.0
4                   1.0                        0.0

```

[5 rows x 30 columns]

```
In [20]: df_scaled.to_csv("fraud_final_preprocessed.csv", index=False)
```